Queue Shift: Updating Classifiers Under a Staffing Budget

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Abstract

When a classifier routes cases to specialized queues, a model update changes both prediction accuracy and the distribution of work. Existing backward-compatibility methods usually count case-level changes or negative flips. Those measures do not identify how much workload must move between queues. I define queue shift as half the L_1 distance between incumbent and proposed queue totals, then choose the batch assignment that maximizes candidate-model probability subject to a queue-shift budget. The problem is a minimum-cost flow and therefore has an integral global optimum. At the movement budget generated by any comparison assignment, including any point on a negative-flip-rate interpolation path, the proposed assignment weakly improves total candidate score. With true conditional probabilities, this is a guarantee for conditional expected accuracy; with estimated scores, a simple error term bounds the possible reversal. Simulations with independently estimated models illustrate the result after an out-of-sample release gate. They are not used to claim a universal effect size. The contribution is an incumbent-relative operational constraint and a matched-budget guarantee, not a new transport solver.

1 The operational problem

Suppose a support organization sends predicted billing tickets to a billing team, technical tickets to a technical team, and so on. It replaces its routing classifier after a new feature or model improves accuracy on later held-out data. The raw update may also change the number of cases sent to each team. Even when total demand is fixed, managers may need to move capacity across queues.

This creates a decision after prediction. The organization has already chosen a better candidate model. It now needs to decide which queue receives each case, balancing the candidate's accuracy signal against a limit on workload redistribution. Retraining the model to reduce negative flips addresses a related concern, but it does not directly control queue totals.

Research on prediction churn and backward-compatible updates measures changes to individual decisions, including cases on which a formerly correct prediction becomes wrong (Milani Fard et al., 2016; Bansal et al., 2019; Srivastava et al., 2020; Träuble et al., 2021; Jiang et al., 2022). That protection can matter when people depend on particular decisions. The staffing problem is different because cases can trade queues while every team receives the same amount of work.

Let $i = 1, \ldots, n$ index cases and $k = 1, \ldots, K$ index queues. The incumbent sends case i to a_i, producing load

$$m_k = \sum_{i=1}^{n} \mathbf{1}\{a_i = k\}.$$

For a proposed assignment z, let $n_k(z)$ be its load in queue k. Define

$$\mathcal{S}(z; m) = \frac{1}{2} \sum_{k=1}^{K} |n_k(z) - m_k| = \sum_{k=1}^{K} [n_k(z) - m_k]_+.$$

The equality follows because old and new loads both sum to n. Queue shift is the smallest number of workload units that must cross queue boundaries to reconcile the two load vectors. A value of zero preserves every queue total, even if individual cases exchange places.

That distinction is why negative-flip rate (NFR) does not identify the staffing cost. Consider three cases with true labels $(1,2,1)$ and incumbent assignments $(1,1,2)$. Compare proposed assignments $(2,2,2)$ and $(2,1,1)$. Each proposal has one correct prediction, one negative flip, one positive flip, and two changed case labels. The first changes queue loads from $(2,1)$ to $(0,3)$, so its queue shift is two. The second keeps loads at $(2,1)$, so its queue shift is zero. Accuracy, NFR, and total label churn are identical, but the staffing implications differ.

This metric is intentionally narrow. It treats a case as one unit of workload and queue totals as the operational state. Applications with different service times, skills, or service-level costs need a richer capacity model. The present result isolates the common case in which predicted class determines a staffed queue.

2 Assignment under a queue-shift budget

Let p_{ik} be the candidate model's probability that case i belongs in queue k. For any assignment z, define its average candidate score as

$$V_p(z) = \frac{1}{n}\sum_{i=1}^{n} p_{i,z_i}.$$

Given an integer movement budget B, Queue Shift solves

$$z^*(B) \in \arg\max_{z} V_p(z) \quad \text{subject to} \quad \mathcal{S}(z;m) \le B. \tag{1}$$

The optimization is a minimum-cost flow. A source sends one unit through each case node. Every case connects to every queue with cost $1 - p_{ik}$. Queue k sends up to m_k units directly to the sink at no movement cost. Additional units pass through one shared overflow node, whose edge to the sink has capacity B. Integer supplies and capacities imply an integral optimum, so the linear program assigns every case to exactly one queue.

The transport construction itself is established. Probability-based label allocation and classification under capacity constraints already use transport or mathematical programming (Tai et al., 2021; Rabkin et al., 2024). More broadly, predict-then-optimize methods incorporate downstream decisions into predictive systems (Elmachtoub and Grigas, 2022). The contribution here is the incumbent-relative queue-shift constraint and the comparison it permits with model-update compatibility rules.

2.1 A matched-budget guarantee

NFR interpolation forms probabilities

$$p^{\alpha}_{ik} = (1-\alpha)p^{0}_{ik} + \alpha p^{1}_{ik},$$

where p^0 and p^1 are incumbent and candidate probabilities. Its hard assignment b^{α} traces a path from the incumbent to the raw candidate as α moves from zero to one. This is a strong baseline because a deployment team can select a point that meets an NFR target while retaining validation accuracy. Queue Shift does not need to dispute that selection rule. It takes the resulting movement $B_{\alpha} = \mathcal{S}(b^{\alpha};m)$ as its budget.

Proposition 1 (Dominance at matched queue movement)**.** *For any baseline assignment b, set $B = \mathcal{S}(b; m)$. Then*

$$V_p(z^*(B)) \geq V_p(b).$$

Proof. The baseline b is feasible for Equation 1 by the definition of B. An optimum cannot have lower objective value than a feasible comparison assignment. □

The proposition applies to every NFR interpolation point and, in fact, to any update assignment. It is not a simulation result. The raw candidate endpoint yields equality in candidate score because choosing the highest-probability queue separately for every case is unconstrained-score optimal. At budget zero, however, Queue Shift can exchange cases while preserving all incumbent queue totals.

If p_{ik} is the true conditional probability of class k, then $V_p(z)$ is conditional expected accuracy. The proposition therefore gives an expected-accuracy guarantee. Estimated probabilities require more care. Let q_{ik} denote the true conditional probabilities and $e_{ik} = q_{ik} - p_{ik}$. For $z^* = z^*(B)$ and baseline b,

$$V_q(z^*) - V_q(b) \geq V_p(z^*) - V_p(b) - \frac{1}{n} \sum_{i=1}^{n} \left(|e_{i,z_i^*}| + |e_{i,b_i}| \right). \tag{2}$$

Thus score dominance is exact, while accuracy dominance depends on probability quality. A uniform error bound $|e_{ik}| \leq \varepsilon$ simplifies the final term to 2ε. The bound is conservative, but it states precisely what a deployment validation must establish.

3 Estimated-model validation

The simulation asks whether the matched-budget advantage remains visible when both classifiers are fitted from finite historical data. It does not estimate an effect for real support teams. The full design is recorded in `validation_plan.md`, and every reported result is generated from checked-in seeded outputs.

Five queues have unequal base rates. Labels and features come from a fixed Gaussian population. The incumbent is a multinomial logistic regression fitted on one signal block. The candidate is fitted on that block plus an independent innovation block. An update enters the planning exercise only if the raw candidate is more accurate on an independent sample of 2,000 labeled cases. This sample can be interpreted as out-of-time release data when the data-generating process is stable. The later planning batches are not labeled when assignments are chosen.

The scenarios vary historical sample size, innovation strength, and candidate-score temperature. The last perturbation changes the scores used in Equation 1 without changing hard candidate predictions. Each scenario contains 50 independently trained model pairs, four future batches per accepted pair, and 300 cases per batch. Uncertainty is computed across model pairs after averaging their planning batches.

Figure 1 compares Queue Shift with NFR interpolation at exactly the same realized queue movement. At $\alpha = 0.5$, mean conditional expected gains range from 0.15 to 3.03 percentage points across the five scenarios. Mean realized gains range from 0.18 to 2.95 points. The weakest case has a realized 95% interval that crosses zero. At $\alpha = 0$, preserving all incumbent queue totals still permits gains because cases can exchange queue slots. Candidate updates pass the release gate in at least 88% of trained pairs.

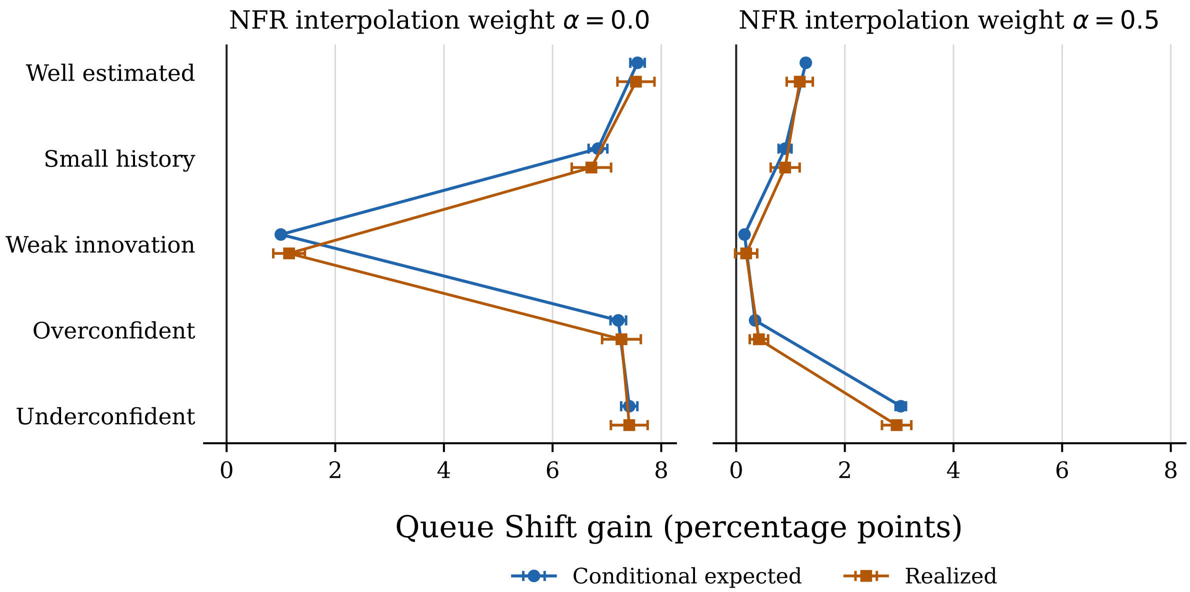

Figure 1: Queue Shift gain over NFR interpolation at matched queue movement. Points are means across independently trained, accepted model updates; bars are 95% normal intervals. Conditional expected accuracy uses the simulation's true probabilities and is not observable in deployment.

The stress test also shows the limit of the claim. Under weak innovation at $\alpha = 0.75$, the mean conditional expected gain is approximately zero and slightly negative before rounding. At $\alpha = 1$, all scenarios satisfy the required score and accuracy equality. These results are consistent with Equation 2: optimizing noisy probabilities can lose a small amount of true value when the available score advantage is nearly exhausted.

4 What a deployment would validate

A practical rollout separates model release from batch assignment. First, compare the raw candidate with the incumbent on later labeled data and reject candidates that do not improve the chosen predictive criterion. Second, choose an operational queue-shift budget. An existing NFR rule can supply that budget, or managers can state it directly. Third, solve Equation 1 on each planning batch using candidate probabilities. Finally, monitor queue loads by construction and evaluate accuracy when labels arrive.

The mathematical guarantee needs no labels at assignment time, but the expected-accuracy interpretation needs useful probability scores. Reliability plots, proper scoring rules, and out-of-time replay can diagnose that condition. Equation 2 also suggests a direct sensitivity analysis: perturb candidate scores within an empirically plausible error set and report how often the chosen assignment changes or loses value.

Three limitations matter before public claims become broad. First, the experiments assume a stable population. They test a model innovation, not demand drift. Second, queue shift measures net load redistribution, not the identities of workers or cases. Third, the evidence is synthetic. A field study should report service times, queue-specific capacity, forecast horizon, and realized staffing actions. Until then, the supported claim is the optimization guarantee and a simulation showing that estimated scores can retain an advantage over the NFR-interpolation baseline.

5 Conclusion

For routing classifiers, compatibility is partly an allocation problem. NFR can remain valuable for protecting individual decisions, but it cannot substitute for a measure of queue totals. Queue Shift makes the operational constraint explicit and optimizes within it. The method weakly dominates any comparison assignment under the common candidate score at the same queue movement, and probability error determines whether that advantage transfers to true accuracy. This result is modest, exact, and testable. It is a better foundation for deployment than treating a simulated headline magnitude as a fact.

A Oracle-posterior experiment

As a mechanical check, an oracle experiment supplies the exact incumbent and candidate posteriors from the same stable Gaussian population. The candidate observes an additional independent signal and is released after an independent accuracy win. Figure 2 shows the full matched movement path. At the zero-movement endpoint, Queue Shift gains 7.64 percentage points; at the raw candidate endpoint, the gain is zero. These magnitudes describe this data-generating process only.

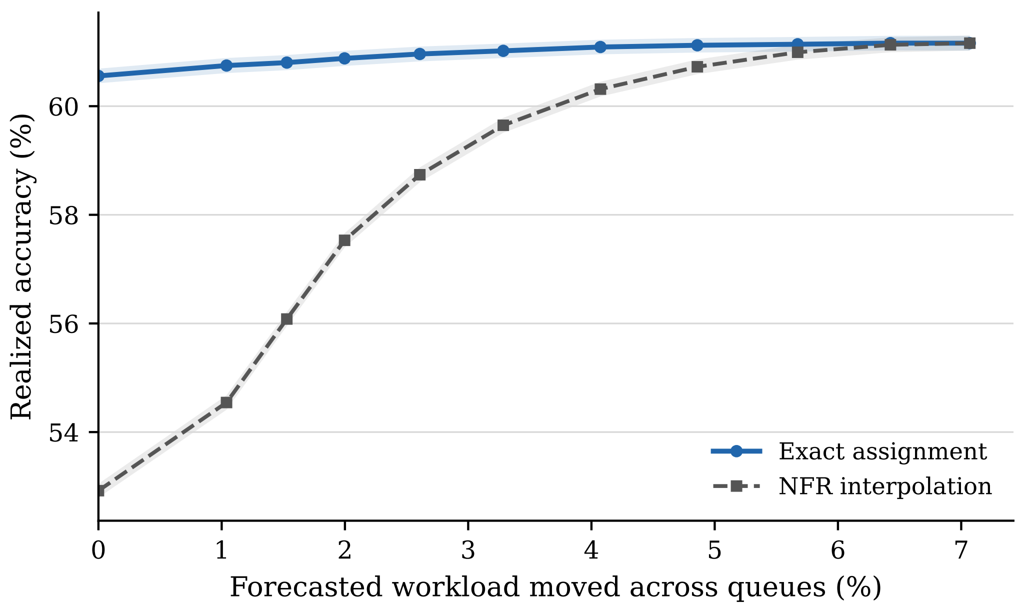

Figure 2: Oracle-posterior comparison at matched queue movement. Bands are 95% normal intervals across independently generated updates.

Table 1: Estimated-model validation against NFR interpolation

Scenario	α	Shift (%)	Score gain (pp)	True gain [95% CI] (pp)	Realized gain [95% CI] (pp)
Well estimated	0.0	0.00	7.75	7.56 [7.43, 7.70]	7.54 [7.19, 7.88]
Well estimated	0.5	3.82	1.35	1.28 [1.23, 1.33]	1.17 [0.93, 1.41]
Well estimated	1.0	7.42	0.00	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Small history	0.0	0.00	8.53	6.84 [6.66, 7.01]	6.71 [6.35, 7.08]
Small history	0.5	4.11	1.48	0.90 [0.78, 1.02]	0.90 [0.64, 1.17]
Small history	1.0	7.88	0.00	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Weak innovation	0.0	0.00	1.40	1.00 [0.96, 1.04]	1.15 [0.86, 1.44]
Weak innovation	0.5	1.89	0.33	0.15 [0.13, 0.18]	0.18 [-0.02, 0.39]
Weak innovation	1.0	3.02	0.00	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Overconfident	0.0	0.00	13.30	7.21 [7.07, 7.35]	7.27 [6.91, 7.62]
Overconfident	0.5	5.42	0.91	0.35 [0.31, 0.38]	0.42 [0.25, 0.59]
Overconfident	1.0	7.57	0.00	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Underconfident	0.0	0.00	3.60	7.41 [7.26, 7.56]	7.41 [7.07, 7.75]
Underconfident	0.5	2.76	1.42	3.03 [2.93, 3.13]	2.95 [2.68, 3.22]
Underconfident	1.0	7.45	0.00	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]

Note: Multinomial logistic models are estimated on historical samples and released only after the candidate wins on an independent labeled sample. Each row compares exact assignment with probability interpolation at the same realized queue shift. True gain uses the simulation's conditional probabilities and is unavailable in applications. Results average four planning batches within each independently trained accepted update before averaging across updates.

Table 2: NFR interpolation versus exact assignment at the same queue shift

Interpolation α	Queue shift (%)	NFR (%)	NFR path accuracy (%)	Exact assignment accuracy (%)	Gain (pp)
0.0	0.00	0.00	52.92	60.56	7.64
0.2	1.53	2.30	56.08	60.80	4.72
0.5	3.28	6.49	59.65	61.02	1.37
0.8	5.67	10.79	60.99	61.14	0.15
1.0	7.07	13.28	61.16	61.16	0.00

Note: The data-generating process is fixed across release and planning samples. The candidate observes an additional independent signal and is deployed only after beating the incumbent on an independent out-of-sample release set. Each row compares probability interpolation with the exact assignment using the candidate posterior at the interpolation point's realized queue shift. Results average 1,000 planning batches from 200 accepted updates. Percentage-point gains use labels revealed only after both assignments are fixed.

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